Evidence-Bounded Stochastic Petri Net Control in SCPN-Fusion-Core: Reduced-Order Results and Open Validation Gaps

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Abstract

SCPN-Fusion-Core is an open-source Python–Rust research toolkit combining axisymmetric equilibrium, reduced-order transport and control experiments. We report an evidence-bounded snapshot of its stochastic-Petri-net control path at software commit `598beb237e78`; the full revision is recorded in the paper metadata. The production compiler maps each Petri-net transition to one threshold-configured leaky integrate-and-fire unit when the optional `sc_neurocore` backend is available, and retains a deterministic float compatibility path. Continuous markings are clipped to $[0, 1]$ and actuator readout is magnitude- and slew-bounded. This source-level compiler contract is distinct from the vertical-control experiment below: that benchmark compares the implemented PID, `LQRController` and `SpikingControllerPool` directly and does not execute a compiled Petri artifact.

On a deterministic reduced-order vertical-instability model with growth rate $200\,\mathrm{s}^{-1}$, a 5 mm disturbance and a 1 kHz update, PID and LQR settle in 25 ms and 39 ms, respectively. The spiking controller does not stabilise the case, reaches 159.683 mm peak displacement and has no settling time. Its internal dynamics lag the 5 ms instability e-folding time. This negative result rules out a fast-VDE performance claim for the current SNN.

The equilibrium evidence is similarly bounded. A provenance-bound same-case, warm-start reproduction of public DIII-D EFIT shot 145419 at 2100 ms reaches 1.908% deep-region RMS poloidal-flux error after 21 accelerated iterations, whereas a plasma-free cold start remains at 126.788% deep RMS and cannot select the physical branch. These are reproduction and failure-characterisation results, not blind prediction. The current pointwise SPARC artifact records 16/16 gated rows on the neural backend, but only 8/16 also pass the adapted source contract. The pointwise gate does not supply the same-case external-coil description required for predictive free-boundary reconstruction. We therefore make no claim of experimental plasma control, production PCS readiness, portable hardware speed, or free-boundary SPARC prediction.

The contribution is the auditable boundary itself: implemented neuro-symbolic software, deterministic reduced-order experiments, provenance-carrying evidence, and explicit failure semantics. Required next steps are independently sourced coil and diagnostic inputs, shot- and device-disjoint validation, calibrated uncertainty and out-of-domain detection, hardware synthesis and measurement, and fair same-case comparisons against established control and equilibrium workflows.

Keywords: stochastic Petri net; spiking neural network; tokamak control; Grad–Shafranov; reduced-order validation; negative result

Software version and date

This preprint describes `scpn-fusion` release 4.1.0. The bound commit `598beb237e78` of 12 November 2026 is part of that release, and the seven implementation sources named in the evidence manifest are byte-identical at the release tag `v4.1.0`. On 13 November 2026 the vertical benchmark, the DIII-D reproduction and the SPARC pointwise benchmark were rerun on the release line and reproduced every number above. Compared with the review draft, only this section, the date and the author affiliation changed.

Methods and evidence boundary

The Petri-net formulation follows the standard bipartite place–transition structure [Petri, 1962, Murata, 1989]; the spiking paths use leaky integrate-and-fire dynamics [Maass, 1997]. The compiler and vertical controller are two separately audited software surfaces. No evidence in this package establishes an end-to-end Petri-to-spiking vertical-control execution. The vertical benchmark is a deterministic ordinary-differential-equation simulation, not a plasma discharge. Hardware latency and radiation response have not been measured.

The DIII-D EFIT reference [Lao et al., 1985] is distributed through the public OMAS sample data. The equilibrium reproduction uses reference-derived flux anchors and a measured external-source description. The cold-start failure demonstrates why this cannot be described as independent reconstruction. SPARC reference files [Creely et al., 2020] remain useful test inputs, but their presence does not establish experimental accuracy for every subsystem.

The package carries exact byte copies of the three cited JSON records, their source revisions, Git blob identities and SHA-256 digests in `evidence/evidence_manifest.json`. The deterministic generator validates the reported scalars and the implementation-source digests. These checks repackage existing results; no fresh discharge, equilibrium or controller run was performed for this draft.

Availability and status

Code is available at `https://github.com/anulum/scpn-fusion-core`. This is a venue-neutral review draft and has not been submitted to a venue. No DOI has been reserved and the manuscript-content licence has not yet been selected by the author. Reproduction instructions and the machine-readable claim boundary are included in this submission directory.

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